International Islamic University on Chittagong
Center for General Education (CGED)
Midterm Examination-Spring-2025

Course Title: Life and Teachings of Prophet (SAAS)
Course Code: URED-3604

Time 1:30 Hours **Full Marks: 30**

Answer all questions strictly

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	A) What do you understand by *Sirah*? Discuss the benefit of studying the *Sirah* of the Prophet (SAAS) in the light of the Qur'an and *Sunnah*. Or, B) What is *Ayyam Jahiliyyah*? Give a brief socio-political background of *Ayyam Jahiliyyah*.	10	2	Remember & Understand
2	Why did the disbelievers of Makkah turn against the Prophet Muhammad (SAAS) despite their testimony to him as *al-Ameen* (most trustworthy) and *al-Sadique* (most thruthful)? What were the attempts taken by the Quraiysh to stop him from his missions?	10	2	Understand
3	Write short notes on the following topics: (Any two) a. Boycott and confinement b. *Isra'* and *Mi'raj* c. Journey to *Ta'if*	10	2	Remember

International Islamic University Chittagong
Center for General Education (CGED)

Midterm Examination: Spring 2025
Course Code: GEHE-3601

Time: 1 hours and 30 minutes.

Program: Undergraduate
Course Title: History of Emergence of of Bangladesh
Full Marks: 30

Instructions:
i. All Questions are Compulsory.
ii. Figures in the right margin indicate full marks.
iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.						
Letter of Symbol	R	U	App	An	E	
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Cr

	Text of the Questions	Marks	Bloom's Level	CLO
1	Sketch out the Physiographic features of Bangladesh and explain the geographical impacts on the life and society of the country.	10	An	CLO1
2	Elucidate the political episodes and factors that led to the emergence of Pakistan in 1947 on the basis of Two Nation Theory. Or Explain the Socio-political context for the 'United Independent Bengal' movement in 1947 and analyze the reasons of its failure.	10	An	CLO2
3	Review the historical background of the Language Movement of 1952 and evaluate its significances in the development of Bangla Language, Literature and Culture.	10	E	CLO2

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. Engineering in CSE

Midterm Examination, Spring'25

Course Code: ECON-3501 | Course Title: Principles of Economics

Time: 1 hour 30 minutes | Full Marks: 30

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.

(ii) Course Outcomes (COs) and Bloom's Levels are mentioned in additional Columns.

Course Outcomes (COs) of the Questions	
CO1	Explain the knowledge of the fundamental concepts and theories of micro and macro-economics.
CO2	Analyze the key indicators of economic growth.
CO3	Comparethe economic theories and concepts to analyze behavior of individuals, firms and nations to act as a responsible citizen.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

1)	a)	Economics is all about production, distribution and consumption. Do you agree? Explain your viewpoint in the light of modern definition of economics.	CO1	U	5
1)	b)	Illustrate the differences between micro and macroeconomics.	CO1	An	5
2)	a)	What do you mean by elasticity of demand? Explain different types of Elasticity of demand.	CO2	R	5
2)	b)	Consider the following demand and supply functions of widgets $P = 80 - Q$ (Demand), $P = 26 + 2Q$ (Supply). Find the equilibrium price and quantity. Show the equilibrium position in a graph.	CO2	Ap	5
3)	a)	What does 'Production' mean in economics? Define the 3 basic factors of production.	CO3	U	5
3)	b)	What do you mean by 'variable production input' and 'fixed production input'? Describe these two types of inputs with an example.	CO3	E	5
		OR			
3)	a)	What is returns to scale? Discuss the various types of returns to scale.	CO3	U	5
3)	b)	Write short notes on: Technological impact on production, Barter transection.	CO3	E	5

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

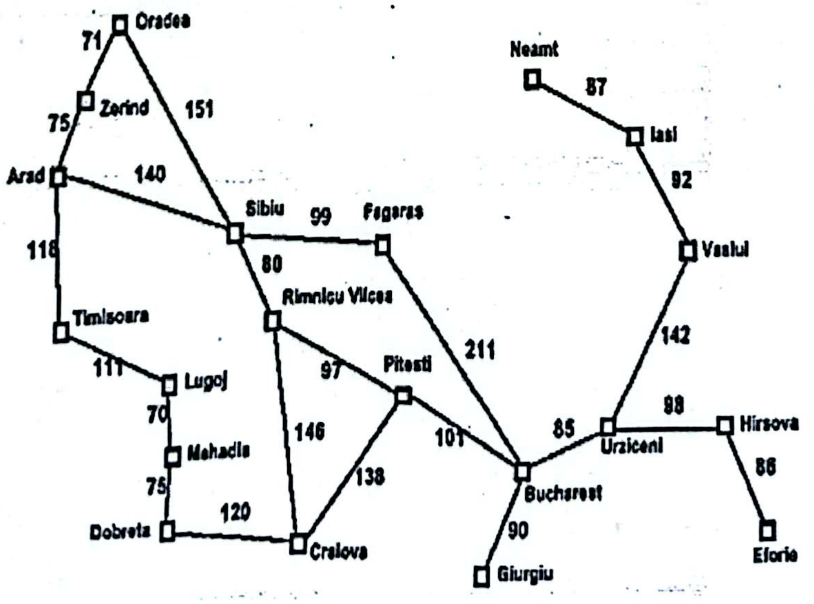
B. Sc. in CSE Semester Midterm Examination, Spring-2025

Course Code: CSE-3635 Course Title: Artificial Intelligence

Total marks: *30* Time: *1* hour *30* minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks]

			CLO	Marks
1.	a)	Define in your own words: i. artificial intelligence ii. rationality	CLO1	1
1.	b)	How can you express the goals of rational agents? Explain with example.	CLO2	2
1.	c)	Suppose you are designing an autonomous rescue robot. The objective of the robot is to navigate flood-affected areas, identify trapped individuals, and assist in safe evacuation or alert rescue teams. i. Write down the PEAS specification for the robot. ii. Characterize the robot's environment with appropriate reasoning as fully observable vs. partially observable, deterministic vs stochastic, episodic vs sequential, discrete vs continuous. iii. From the following types of agents, select the most suitable structure for designing the rescue robot with proper explanation. a. Simple Reflex Agents b. Model-Based Reflex Agents c. Goal-Based Agents d. Utility-Based Agents e. Learning Agents	CLO3	2+3+2
2.	a)	What do you mean by uninformed search? Discuss the limitations of the simple hill climbing algorithm.	CLO1	2
2.	b)	Define heuristic. Write down the heuristics of 8-queens problem.	CLO2	2
2.	c)	(see below)	CLO4	3



Straight-line distance to Bucharest	
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Assume that the initial state is Arad and the goal state is Bucharest. Show h[illegible] search strategy would create a search tree to find a path from initial s[illegible] state.

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

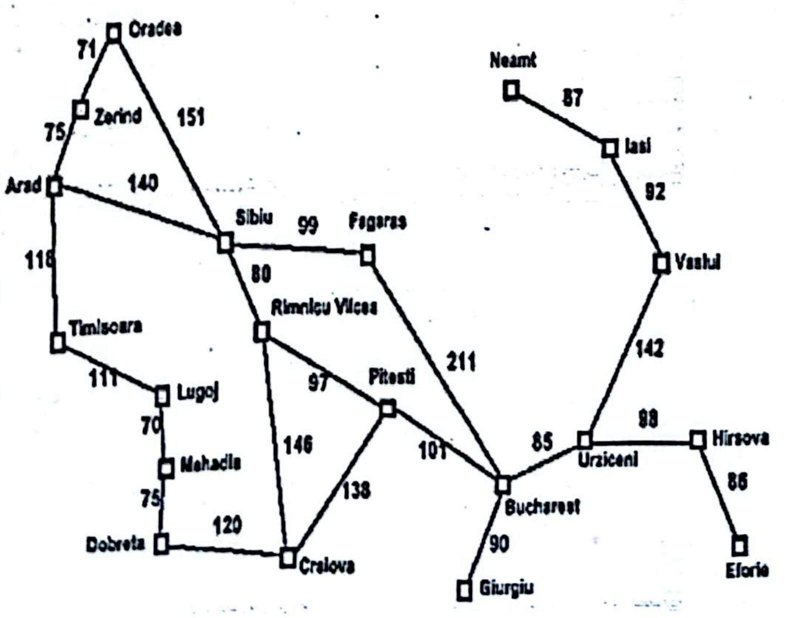
B. Sc. in CSE Semester Midterm Examination, Spring-2025

Course Code: CSE-3635 Course Title: Artificial Intelligence

Total marks: *30* Time: *1* hour *30* minutes

[Answer *all* the questions. Figures in the right-hand margin indicate full marks]

			CLO	Marks
1.	a)	Define in your own words: i. artificial intelligence ii. rationality	CLO1	1
1.	b)	How can you express the goals of rational agents? Explain with example.	CLO2	2
1.	c)	Suppose you are designing an autonomous rescue robot. The objective of the robot is to navigate flood-affected areas, identify trapped individuals, and assist in safe evacuation or alert rescue teams. i. Write down the PEAS specification for the robot. ii. Characterize the robot's environment with appropriate reasoning as fully observable vs. partially observable, deterministic vs stochastic, episodic vs sequential, discrete vs continuous. iii. From the following types of agents, select the most suitable structure for designing the rescue robot with proper explanation. a. Simple Reflex Agents b. Model-Based Reflex Agents c. Goal-Based Agents d. Utility-Based Agents e. Learning Agents	CLO3	2+3+2
2.	a)	What do you mean by uninformed search? Discuss the limitations of the simple hill climbing algorithm.	CLO1	2
2.	b)	Define heuristic. Write down the heuristics of 8-queens problem.	CLO2	2
2.	c)		CLO4	3



Straight-line distance to Bucharest

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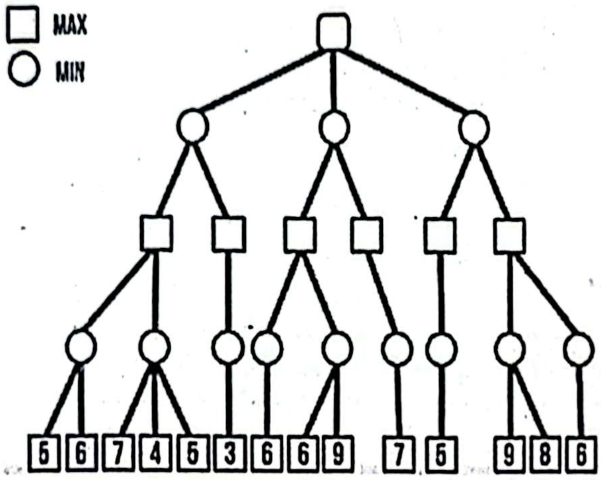
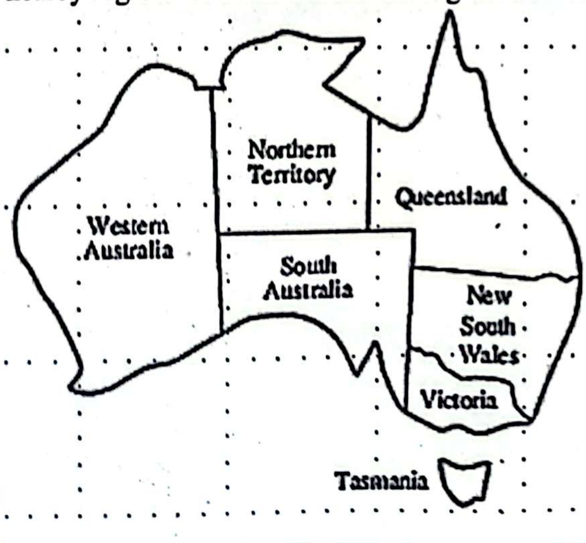
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Assume that the initial state is Arad and the goal state is Bucharest. Show how A* search strategy would create a search tree to find a path from initial state to goal state.

2.	d)	Discuss how genetic algorithms work in contrast to biological concepts?	CLO2	3
3	a)	Define CSP. What are the differences between Search-Based Problem and CSP?	CLO 1	1+2
3	b)	Consider the following two player game tree in which the static scores are given from the first players point of view. Apply the alpha beta pruning and compute the value of the root of the tree. Also find the most convenient path for MAX node.  	CLO3	3
3	c)	Consider the Australian map below. You want to color areas so that none of their neighbors have the same color. The colors you may choose from are Red, Blue and Green. Start from western Australia. When you are in an area, you may go to nearby regions or color the current region. Formulate the problem.  	CLO2	4

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Midterm Examination — Spring 2025

Course Code: CSE 3525 — Course Title: Data Communication

Total marks: 30 — Time: 1 hour 30 minutes

		M	C	D
1.				
a)	Describe the functions of each of the layers of ISO-OSI model.	5	1	2
b)	Discuss encapsulation, decapsulation, and addressing at the different layers of the TCP/IP model. Include necessary diagrams to support your explanation.	5	1	1
2.				
a)	What are the propagation time, transmission time, and total latency for a 5 MB message (an image) if the network bandwidth is 1 Mbps? Assume the distance between the sender and receiver is 9,600 km, the speed of light in the medium is 2.4×10^8 m/s, the queuing delay is 10 ms, and the processing delay is 20 ms.	5	2	3
b)	Assume you have a channel with a bandwidth of 3 MHz and an SNR of 63. What are the appropriate bit rate and signal level for this channel?	5	2	3
3.				
a)	What do you mean by transmission impairments? How can transmission impairments be mitigated?	5	2	3
b)	If the loss of a cable is labeled as -1.2 dB/km, what will be the power at point P2 in the following link? Assume that y is the last digit of your ID, and the power at point P1 is 2y mW, as shown in the figure below. 	5	2	3
	Or			
3.				
a)	Construct the digital signals for the following bit stream: 100100000011, using the following line coding schemes: i. NRZ-I ii. Manchester iii. Differential Manchester iv. AMI v. AMI with HDB3.	5	2	4
b)	Explain how fiber optic cables transmit light signals.	5	2	2

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid Exam, Spring 2025

Course Code: CSE 3631 **Course Title:** Operating System

Time: 1 hours 30 minutes **Full Marks:** 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes are mentioned in additional Columns

[Answer the questions from the followings]

1. a) How does the distinction between kernel mode and user mode function as a rudimentary form of protection (security)? CO1 03

   b) Explain the layered approach to operating system design. What are the advantages and disadvantages of using this approach? CO1 04

   c) What are some advantages of peer-to-peer systems over client–server systems? CO1 03

2. a) In a mobile operating system (like Android), background and foreground tasks are managed using a hybrid scheduling algorithm. High-priority tasks (e.g., UI rendering, audio playback) are given preemptive priority over background services (e.g., syncing data, notifications). However, to ensure fairness among tasks of the same priority (e.g., multiple background apps), Round Robin is used. CO2 07

   The system uses: Three priority levels: High (UI/audio), Medium (background apps), Low (background daemons) and RR time quantum: 10 ms for each level.

   Consider the following scenario :

Task	Arrival	Burst Time	Priority	Description
UI Thread	0 ms	30 ms	High	Touch input + UI rendering
Sync App	5 ms	20 ms	Medium	Google Drive background sync
Music App	7 ms	25 ms	High	Audio playback
Cleanup Daemon	10 ms	15 ms	Low	Periodic log cleanup
Chat Notification	12 ms	10 ms	Medium	Push notification background task

Now, answer the following questions :

1. Simulate the execution(Gantt Chart) of these tasks for using hybrid Priority + RR scheduling (RR quantum = 10 ms).

2. Calculate the average waiting time.

3. In the scenario above, identify potential starvation risks for the low-priority background daemon. Explain how the hybrid RR + Priority approach contributes to or mitigates starvation.

b) Describe the actions taken by a kernel to context-switch between processes. C01 03

3. a) Explain the concept of Process Control Block data structure, that is maintained by the operating system to keep track of a particular process. CO2 02

b) A cloud computing platform is processing multiple tasks with varying burst times. The server uses Shortest Remaining Time First (SRTF) scheduling to ensure that tasks with the shortest remaining burst time are processed first. The tasks are submitted at different times, and the platform needs to decide how to schedule them. CO2 05

Task	Arrival Time (ms)	Burst Time (ms)
T1	0	12
T2	2	8
T3	4	6
T4	5	10

Simulate the execution of these tasks using SRTF scheduling.

- Draw the Gantt chart.
- Calculate the waiting time and turnaround time for each task.
- Analyze how preemption works in SRTF when new tasks arrive while others are still executing. Which task experiences the most preemption, and why?

c) A drone navigation system uses EDF to manage tasks related to sensor updates, trajectory computation, and battery management. C02 03

- **T1 (Sensor update):** Execution = 1 ms, Relative Deadline = 5 ms
- **T2 (Trajectory computation):** Execution = 3 ms, Relative Deadline = 10 ms
- **T3 (Battery check):** Execution = 1 ms, Relative Deadline = 8 ms

1. Simulate the execution of these tasks over a 20ms interval using EDF. Analyze how deadlines are prioritized and whether any tasks miss their deadlines.
2. Compare the EDF schedule with a Rate Monotonic schedule for the same task set. Which scheduling policy better suits this drone application, and why?

International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B. Sc (Engg.) in CSE **Semester: 6th**

Mid-Term Examination, Spring 2025

Course Code: **CSE-3637** Course Title: **Software Engineering**

Time: 1 hour and 30 minutes **Total Marks:** 30

[Answer all the questions. Figures in the right-hand margin indicate full marks.]

			Marks	CO	DL
Q.1.	**a.**	Read the passage and answer questions I and II: Suppose you are developing a complex software system, and you want to break that software system into smaller, more manageable components and design these components to be independent and interchangeable components. I. Which software engineering principle will you follow to meet up the above requirement? Use an example to explain the principle. II. Are there any other software principles that will make software more robust and manageable? Briefly explain them.	5	CO2	C2
	b.	What is a framework? What programming languages are you familiar with?	2	CO1	C1
	c.	What are the latest trends in modern software engineering? Explain any two emerging technologies used in software development and their impact on the industry.	3	CO1	C2
Q.2.	**a.**	List down some segments of software cost. Do you think software maintenance is more costly than building new software? Justify your answer.	4	CO2	C1
	b.	If you were a project manager using the waterfall model, how would you handle a situation where a client requests a significant change to the project scope during the testing phase? What steps would you take to ensure that the change is implemented smoothly and without causing delays or compromising the project's overall quality?	4	CO2	C2
	c.	"Hardware wears out, but software changes." Do you agree with this statement? Justify your answer. **Or** What is Software? What is generic and custom-made software? Give examples.	2	CO1	C1
Q.3	**a.**	A software company is developing an AI-based healthcare monitoring system that requires continuous updates based on patient feedback. Which software process model would be most suitable, and why?	4	CO3	C3
	b.	Compare and contrast the Waterfall Model and the Incremental Model in terms of flexibility, risk management, and real-world applications. **Or** What is Parallel Waterfall Development? Describe a flaw of this methodology	3	CO1	C1
	c.	Differentiate between Engineering and Software Engineering. Provide at least three key differences and relevant real-world examples.	3	CO1	C2